PYTHON DEVELOPMENT

Learn Python. Build Applications. Develop Practical Technology Skills.

Aayush Institute of Professional Studies offers career-oriented **Python Development training** for students, graduates, freshers, working professionals and aspiring developers who want to build strong programming and application development skills.

Python is widely used across software development, web development, automation, data, artificial intelligence and backend application development.

The programme focuses on **Python programming, problem solving, object-oriented programming, database connectivity, web development, APIs, automation and practical project development**.

Depending on the selected programme, learners can progress from **Python Fundamentals to Python Development or Python Full Stack Development**.

WHAT IS PYTHON DEVELOPMENT?

Python Development involves using the Python programming language to create software, web applications, automation solutions, APIs and other technology products.

A Python development pathway can include:

Programming → OOP → Database → Backend → APIs → Web Development → Testing → Deployment → Projects

Python can also serve as a foundation for advanced learning in:

- Data Analytics
- Artificial Intelligence
- Machine Learning
- Generative AI
- Automation
- Data Science

WHO CAN LEARN PYTHON?

Python can be suitable for:

- 10th / 12th Pass Students
- ITI Students
- Polytechnic / Diploma Students
- BCA Students
- B.Sc. IT / Computer Science Students
- B.Tech / B.E. Students
- MCA Students
- M.Tech Students
- Graduates
- Freshers
- Working Professionals
- Existing Developers
- Data / AI Aspirants
- Freelancers
- Career Changers

Beginners can start with Python fundamentals without requiring prior programming experience.

PYTHON DEVELOPMENT LEARNING PATH

01. COMPUTER & PROGRAMMING FUNDAMENTALS

Build the foundation required for software development.

Topics Include:

- Computer Fundamentals
- Operating System Basics
- Internet Fundamentals
- Software Concepts

- Development Environment
- Command Line Fundamentals
- Programming Concepts
- Problem Solving
- Algorithms & Logic

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02. PYTHON FUNDAMENTALS

Learn the core concepts of Python programming.

Topics Include:

- Python Installation & Setup
- Python Syntax
- Variables
- Data Types
- Operators
- Input & Output
- Conditional Statements
- Loops
- Functions
- Lists
- Tuples
- Sets
- Dictionaries
- Strings
- Modules
- Packages

- File Handling
- Exception Handling

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03. OBJECT-ORIENTED PROGRAMMING

Understand how Python applications can be structured using object-oriented concepts.

Topics Include:

- Classes
- Objects
- Constructors
- Methods
- Encapsulation
- Inheritance
- Polymorphism
- Abstraction
- OOP-Based Application Design

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04. ADVANCED PYTHON CONCEPTS

Develop stronger programming and application-building skills.

Topics May Include:

- List & Dictionary Comprehensions
- Lambda Functions
- Iterators
- Generators
- Decorators
- Regular Expressions

- Virtual Environments
- Package Management
- Working with External Libraries
- Advanced Exception Handling
- JSON Processing
- Data Handling

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05. FILE & DATA PROCESSING

Learn how Python can work with different types of data and files.

Topics Include:

- Text Files
- CSV Files
- JSON
- File & Folder Operations
- Data Import / Export
- Data Processing
- Basic Automation
- Structured Data Handling

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06. DATABASE WITH PYTHON

Learn how Python applications communicate with databases.

Topics May Include:

- Database Fundamentals
- SQL Fundamentals
- MySQL

- PostgreSQL
- SQLite
- Database Connectivity
- CRUD Operations
- Queries
- Tables & Relationships
- Python Database Integration

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07. PYTHON WEB DEVELOPMENT

Learn how Python can be used to develop web applications.

DJANGO

Topics may include:

- Django Fundamentals
- Project Structure
- Applications
- URL Routing
- Views
- Templates
- Models
- Forms
- Database Integration
- Authentication
- Admin Panel
- CRUD Applications
- REST API Fundamentals

FLASK

Topics may include:

- Flask Fundamentals
- Application Structure
- Routing
- Templates
- Forms
- Database Connectivity
- REST APIs
- Authentication Fundamentals
- Application Development

The framework covered depends on the selected programme.

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08. REST API DEVELOPMENT

Learn how Python applications communicate with websites, mobile applications and other services.

Topics Include:

- API Fundamentals
- REST Architecture
- HTTP Methods
- GET / POST / PUT / PATCH / DELETE
- JSON
- Request & Response
- API Authentication
- API Validation
- Error Handling

- Third-Party API Integration
- API Testing

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09. AUTHENTICATION & SECURITY FUNDAMENTALS

Understand the basic concepts involved in protecting web applications and APIs.

Topics May Include:

- User Registration
- Login / Logout
- Authentication
- Authorization
- Password Security
- Sessions / Tokens
- Access Control
- Input Validation
- Basic Web Security Practices

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10. PYTHON AUTOMATION

Python can be used to automate repetitive digital tasks.

Topics May Include:

- File Automation
- Folder Management
- Data Processing
- Excel Automation
- CSV Processing
- Report Generation

- Basic Web Automation
- Task Automation
- Email Automation Fundamentals
- API-Based Automation

Automation projects depend on the selected programme and available tools.

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11. GIT & GITHUB

Learn professional source-code management practices.

Topics Include:

- Git Fundamentals
- Repository Creation
- Commit
- Branch
- Merge
- GitHub
- Version Control
- Project Collaboration
- Code Repository Management

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12. TESTING & DEBUGGING

Learn how to identify and resolve problems in Python applications.

Topics Include:

- Debugging Fundamentals
- Error Handling
- Code Testing

- Application Testing
- Unit Testing Fundamentals
- API Testing
- Log Analysis
- Bug Identification
- Bug Fixing

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13. DEPLOYMENT FUNDAMENTALS

Understand how Python applications can be prepared for deployment.

Topics May Include:

- Application Configuration
- Environment Variables
- Virtual Environments
- Server Fundamentals
- Hosting Concepts
- Web Application Deployment
- API Deployment
- Database Configuration
- Basic Cloud Deployment Concepts

PYTHON FULL STACK DEVELOPMENT

Learners looking for complete web application development can progress from Python into a full stack pathway.

Frontend

HTML5 + CSS3 + JavaScript

Backend

Python + Django / Flask

Database

MySQL / PostgreSQL / SQLite

API

REST API + JSON

Development

Git + GitHub + Testing + Debugging

Deployment

Hosting + Application Deployment Fundamentals

This pathway helps learners understand both the user-facing and server-side components of a web application.

PRACTICAL PYTHON PROJECTS

BEGINNER PROJECTS

- Calculator
- Number Guessing Game
- Quiz Application
- To-Do Application
- Contact Management System
- Student Record Program

INTERMEDIATE PROJECTS

- Expense Management System
- Library Management System
- Student Management System
- Inventory Management System
- Billing Application

- Employee Management System
- File Automation Tool

ADVANCED PROJECTS

- Django Business Application
- E-Commerce Backend
- Appointment Booking System
- CRM Application
- Learning Management Application
- REST API Project
- Business Management System
- Full Stack Web Application

Project complexity depends on the selected programme and learner level.

PYTHON FOR DATA & AI

Python is also commonly used as a foundation for data and AI learning.

Learners interested in these areas can progress into specialised programmes covering:

DATA ANALYTICS

- Python for Data Analysis
- Pandas
- NumPy
- Data Cleaning
- Data Visualisation
- Data Analysis

ARTIFICIAL INTELLIGENCE

- Python for AI

- AI Fundamentals
- Machine Learning Fundamentals
- Generative AI
- AI Tools & Applications

DATA SCIENCE

- Python
- Statistics Fundamentals
- Data Processing
- Machine Learning
- Data Visualisation
- Practical Projects

These areas may be offered as separate specialised courses.

SKILLS YOU CAN DEVELOP

Depending on the selected programme, learners can develop skills in:

- Python Programming
- Programming Logic
- Object-Oriented Programming
- Data Structures Fundamentals
- File Handling
- Data Processing
- SQL
- MySQL / PostgreSQL / SQLite
- Django
- Flask

- REST APIs
- JSON
- Authentication
- Git & GitHub
- Automation
- Testing
- Debugging
- Web Development
- Backend Development
- Full Stack Development
- Project Development

PYTHON CAREER PATHWAYS

Python skills can support entry-level opportunities such as:

- Python Developer
- Junior Python Developer
- Backend Developer
- Django Developer
- Flask Developer
- Python Full Stack Developer
- Software Developer
- API Developer
- Automation Developer
- Application Developer
- Python Development Intern

- Software Development Intern

For data and AI pathways, additional specialised skills and qualifications may be required for roles such as Data Analyst, Data Scientist or Machine Learning Engineer.

Actual job titles, eligibility and employer requirements vary by organisation and role.

FREELANCING & PROJECT WORK

Python skills can also be applied to independent technology projects.

Potential project areas include:

- Business Automation
- Data Processing
- Web Applications
- API Development
- Backend Development
- Database Applications
- Report Automation
- Custom Business Tools
- Website Backends
- Internal Business Applications

Freelancing income and project availability depend on individual skills, portfolio, client requirements and market conditions.

LIVE PROJECT & PRACTICAL EXPOSURE

Depending on the selected programme, learners may receive opportunities for:

- Live Project Training
- Python Application Projects
- Backend Development Projects

- Database Projects
- API Development
- Automation Projects
- Mentor-Guided Development
- Testing & Debugging
- Project Documentation
- Portfolio Development

The objective is to help learners apply programming concepts to practical development requirements.

INTERNSHIP OPPORTUNITIES

Eligible learners may receive information about available internship opportunities in areas such as:

- Python Development
- Web Development
- Backend Development
- Django / Flask
- Software Development
- Automation
- Data / Technology

Paid internships may be available with certain organisations, subject to eligibility, availability and the company's selection process.

Stipend or compensation, where applicable, is determined by the organisation offering the internship.

CAREER GUIDANCE & PLACEMENT SUPPORT

Aayush Institute provides **career guidance at no additional cost** to help learners identify suitable learning and career pathways.

Support may include:

- Career Assessment
- Course Selection Guidance
- Resume / CV Preparation
- Portfolio Guidance
- Technical Interview Preparation
- HR Interview Preparation
- Mock Interviews
- Communication & Workplace Skills
- Job Role Guidance
- Recruitment Opportunity Updates
- Placement Assistance

Placement Information

Placement assistance may be provided to eligible learners based on available opportunities.

Placement is not guaranteed. Employer selection depends on candidate skills, qualification, performance, interview results, vacancies and employer selection criteria.

COURSE LEVELS

PYTHON FOUNDATION

For beginners who want to learn programming.

Programming Fundamentals → Python Basics → Logic Building → Projects

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PROFESSIONAL PYTHON DEVELOPMENT

For learners who want to develop practical Python applications.

Python → OOP → Database → APIs → Automation → Projects

PYTHON WEB DEVELOPMENT

For learners interested in backend and web application development.

Python → Django / Flask → Database → REST APIs → Web Applications

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PYTHON FULL STACK DEVELOPMENT

For learners who want broader web application development skills.

HTML + CSS + JavaScript → Python → Django / Flask → Database → APIs → Git → Deployment → Projects

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YOUR PYTHON DEVELOPMENT JOURNEY

START

Computer & Programming Fundamentals

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LEARN

Python Programming

↓

BUILD

Logic + OOP + Data Handling

↓

CONNECT

Database + SQL + APIs

↓

DEVELOP

Django / Flask + Web Applications

↓

AUTOMATE

Python Automation & Business Tasks

↓

TEST

Debugging + Testing

↓

DEPLOY

Application Deployment

↓

BUILD YOUR PORTFOLIO

Real Projects + Practical Experience + Career Preparation

WHY CHOOSE AAYUSH?

ESTABLISHED SINCE 2002

More than two decades of association with education and professional skill development.

PRACTICAL LEARNING

Focus on applying programming concepts through assignments and projects.

CAREER-ORIENTED TRAINING

Learning pathways are structured around practical technology skills and career preparation.

PROJECT-BASED LEARNING

Learners can work on Python applications and development projects according to their programme level.

CAREER GUIDANCE

Career guidance is provided at no additional cost.

INTERNSHIP OPPORTUNITIES

Eligible learners may receive access to available internship opportunities, subject to organisation requirements and selection.

PLACEMENT SUPPORT

Career preparation and placement assistance are provided based on eligibility and available opportunities.

START WITH PYTHON. BUILD WITH TECHNOLOGY.

Learn programming.
Build applications.
Work with databases and APIs.
Develop practical projects.
Explore web, automation, data and AI pathways.
Build your technology portfolio.

Start Your Python Development Journey with Aayush Institute of Professional Studies.

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